processing. It is located in the cranial cavity and is covered by three cranial meninges. The outer thick layer is **Duramater** which lines the inner surface of the cranial cavity; the median thin layer is **Arachnoid mater** which is separated from the duramater by a narrow **subdural space.** The innermost layer is **Piamater** which is closely adhered to the brain but separated from the arachnoid mater by the **subarachnoid space**. The brain is divided into three major regions: Forebrain, Midbrain and Hindbrain.

Fore Brain

It comprises the following regions: **Cerebrum** and **Diencephalon**. Cerebrum is the 'seat of intelligence' and forms the major part of the brain. The cerebrum consists of an outer cortex, inner medulla and **basal nuclei.** The superficial region of the cerebrum is called **cerebral cortex**, which looks grey due to the presence of unmyelinated nerve cells. Cerebral cortex consists of neuronal cell body, dendrites, associated glial and blood vessels. The surface of the cerebrum shows many convolutions (folds) and grooves. The folds are called **gyri** (singular gyrus); the shallow grooves between the gyri are called **sulci** (singular sulcus) and deep grooves are called fissures. These sulci and gyri increase the surface area of the cerebral cortex. Several sulci divide the cerebrum into eight lobes: a pair of **frontals, parietals, temporals and occipital lobes** (Figure10.7 & Table 10.2).

A median longitudinal fissure divides the cerebrum longitudinally into two cerebral hemispheres (Figure 10.7). A transverse fissure separates the cerebral hemispheres from the cerebellum. The hemispheres are connected by a tract of nerve fibres called **corpus callosum**. Cerebral cortex has three functional areas namely **sensory areas** occur in the parietal, temporal and occipital lobes of the cortex. They receive and interpret the sensory impulses. **Motor area** of the cortex which controls voluntary muscular movements lies in the posterior part of the frontal lobes. The areas other than sensory and motor areas are called **Association areas** that deal with integrative functions such as memory, communications, learning and reasoning. Inner to the cortex is **medulla** which is white in colour and acts as a nerve tract between the cortex and the diencephalon.

Table 10.2 Functions of brain lobes

Structure	Functions
Frontal	Behaviour, Intelligence, Memory, Movement
Parietal	Language, Reading, Sensation
Temporal	Speech, Hearing, Memory
Occipital	Visual processing

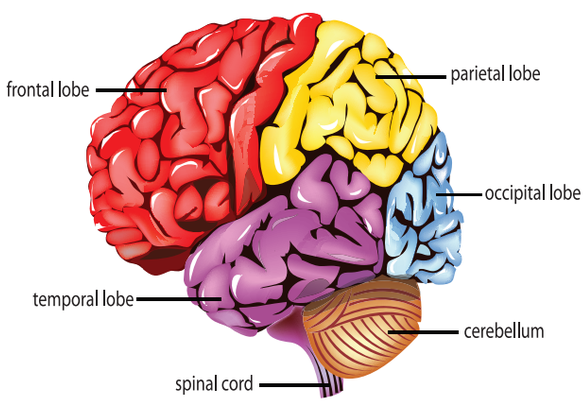


Figure 10.7 Lobes of Cerebral hemisphere

Diencephalon consists largely of following three paired structures.

Epithalamus forms the roof of the diencephalon and it is a non-nervous tissue. The anterior part of epithalamus is vascular and folded to form the anterior **choroid**

TN_GOVT_XI_Zoology_Ch10.indd 183 09-01-2020 5.18.57 PM